UNIT-1 / CHAPTER-1
VECTOR ANALYSIS

Scalars & Vectors:-

A scalar is a quantity which has only magnitude and no direction. e.g. Temparature, Time, Potential etc

A vector is a quantity which has both magnitude as well as direction. e.g. Velocity, Force etc

Vectors are represented by $\vec{A}$ or **A** or by an arrow as $\overrightarrow{A}$.

Scalar & Vector Fields:

A field is a region in which, at each point, there is a corresponding value of some physical function.
There are two types of fields:-

1) Scalar field: If the value of the physical function at each point is a scalar quantity, then, the field is a scalar field.
2) Vector field: If the value of the physical function at each point is a vector quantity, then, the field is a vector field.

Unit Vector:-

A vector whose magnitude is unity, i.e '1', is called unit vector. It serves to specify direction only. It is denoted by a cap on the vector name as $\hat{A}$.

$\Rightarrow \vec{A} = |\vec{A}|\,\hat{A}$ i.e. Vector = Magnitude × Unit Vector

$\Rightarrow \hat{A} = \dfrac{\vec{A}}{|\vec{A}|}$

Vector in its component form can be given as,

$\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$

ADDITION & SUBTRACTION OF VECTORS :-

Two Vectors $\vec{A}$ & $\vec{B}$ can be added or subtracted as given below :-

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$$
$$\vec{B} = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}$$

Addition $\Rightarrow$ $\vec{A} + \vec{B} = (A_x + B_x)\hat{x} + (A_y + B_y)\hat{y} + (A_z + B_z)\hat{z}$

Subtraction $\Rightarrow$ $\vec{A} - \vec{B} = (A_x - B_x)\hat{x} + (A_y - B_y)\hat{y} + (A_z - B_z)\hat{z}$

MULTIPLICATION OF VECTORS :-

Multiplication with a Scalar :-

when a vector is multiplied by a scalar e.g. $\vec{A}$ by s, then only magnitude of vector changes but direction of vector remains same. as,

$$s\vec{A} = sA_x \hat{x} + sA_y \hat{y} + sA_z \hat{z}$$

Multiplication of Two Vectors :-

There are two types of product b/w two vectors -

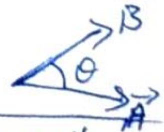
① Scalar Product (Dot Product)

② Vector Product (Cross Product)

① Scalar Product :-

This is known as scalar product because the resultant of this is a scalar quantity.

$$\boxed{\vec{C} = \vec{A}\cdot\vec{B} = \cancel{A}\,|\vec{A}||\vec{B}|\cos\theta = AB\cos\theta}$$



Also $\boxed{\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}}$

where θ = acute angle b/w $\vec{A}$ & $\vec{B}$

Also $\vec{C} = (A_x\hat{x} + A_y\hat{y} + A_z\hat{z})\cdot(B_x\hat{x} + B_y\hat{y} + B_z\hat{z})$

$$= A_xB_x\,\hat{x}\cdot\hat{x} + A_xB_y\,\hat{x}\cdot\hat{y} + A_xB_z\,\hat{x}\cdot\hat{z} + A_yB_x\,\hat{y}\cdot\hat{x}$$
$$+ A_yB_y\,\hat{y}\cdot\hat{y} + A_yB_z\,\hat{y}\cdot\hat{z} + A_zB_x\,\hat{z}\cdot\hat{x} + A_zB_y\,\hat{z}\cdot\hat{y} + A_zB_z\,\hat{z}\cdot\hat{z}$$

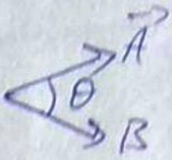
$$\Rightarrow \boxed{\vec{C} = A_xB_x + A_yB_y + A_zB_z}$$

$\hat{x}\cdot\hat{x} = |\hat{x}||\hat{x}|\cos 0 = 1\times1\times1 = 1$

$\hat{x}\cdot\hat{y} = \hat{x}\cdot\hat{z} = \hat{y}\cdot\hat{z}$ --- $= 1\times1\times\cos 90 = 0$

(2) VECTOR PRODUCT :-

The Resultant of this is a vector quantity having magnitude and direction as well. The direction of resultant is Perpendicular direction to plane containing two vectors. The direction can be obtained by right hand screw/palm rule.



According to this, put your Right hand on Ist vector and curl your fingers through acute angle, then thumb will direct in the direction of resultant.

$$\vec{C} = \vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta \, \hat{r}$$

[where θ = acute angle b/w $\vec{A}$ & $\vec{B}$
$\hat{r}$ = unit vector $\perp$ to the plane containing $\vec{A}$ & $\vec{B}$

For $\vec{A} \times \vec{B}$ direction will be inward to the paper & for $\vec{B} \times \vec{A}$ direction is outward of paper i.e.

$$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$$

$$\text{Also, } \vec{C} = (\vec{A} \times \vec{B}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

TRIPPLE PRODUCT :-

(I) Scalar Tripple Product :- Resultant is a scalar

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

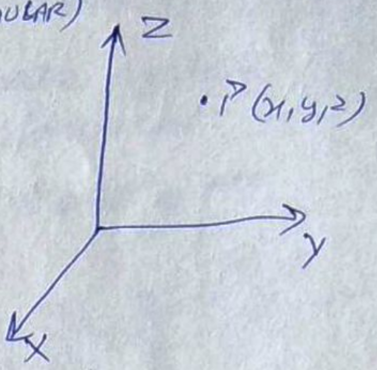
(II) Vector TRIPLE Product :- Resultant is a vector.

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

CO-ORDINATE SYSTEMS :-

① CARTESIAN CO-ORDINATE SYSTEM :- (RECTANGULAR)

In this system, every point have three coordinates namely x, y & z

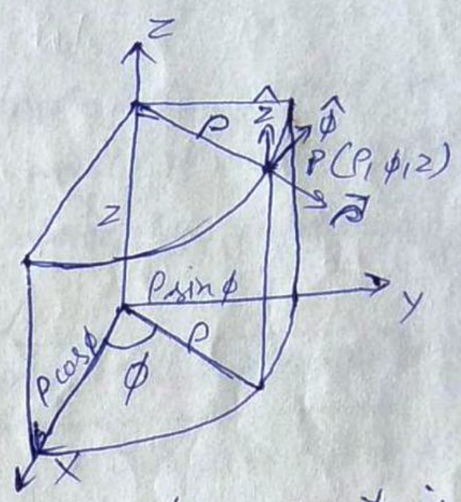


② CYLINDERICAL ~~SPHERICAL~~ CO-ORDINATE SYSTEM :-

In this system, point have co-ordinates namely ρ, ϕ & z

where z is the same as it was in cartesian co-ordinate system.

ρ is the radius of cylinder containing Point P. and about z-axis



and ϕ is the angle b/w XZ plane and the plane containing point P and the Z-axis.

From fig., it is clear that,

$$x = \rho\cos\phi \;,\; y = \rho\sin\phi \;\&\; z = z$$

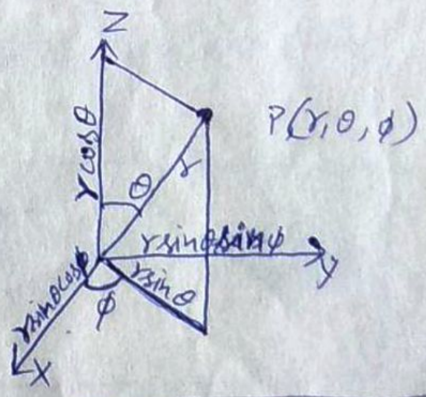
③ SPHERICAL CO-ORDINATE SYSTEM :-

Point have co-ordinates r, θ & ϕ

$r \rightarrow$ Position vector of P

$\theta \rightarrow$ angle b/w z axis & r.

$\phi \rightarrow$ Same as it was in Cylinderical System



From fig.,

$$x = r\sin\theta\cos\phi \;,\; y = r\sin\theta\sin\phi \;\&\; z = r\cos\theta$$

* REMEMBER :-

$x = \rho\cos\phi = r\sin\theta\cos\phi$

$y = \rho\sin\phi = r\sin\theta\sin\phi$

$z = z = r\cos\theta$

$\rho = \sqrt{x^2+y^2} = r\sin\theta$

$r = \sqrt{x^2+y^2+z^2} = \sqrt{\rho^2+z^2}$

$\sin\phi = \frac{y}{\rho}$

$\cos\phi = \frac{x}{\rho}$

$\sin\theta = \frac{\rho}{r}$

$\cos\theta = z/r$

CONVERSION B/W DIFFERENT CO-ORDINATE SYSTEM :-

(I) **Cartesian to Spherical :-** From $A(x, y, z) \rightarrow A(r, \theta, \phi)$

In this a vector in cartesian co-ordinate system is given and it is to be converted in spherical system. i.e. A_x, A_y & A_z are given & A_r, A_θ & A_ϕ are to be obtained

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

As A_x, A_y & A_z will be in x, y & z and A_r, A_θ & A_ϕ should be in terms of r, θ & ϕ, so A_x, A_y & A_z must Ist be converted in terms of r, θ & ϕ. This can be done by replacing x, y & z eqs,

$$\boxed{x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi \ \&\ z = r\cos\theta}$$

(II) **Spherical to cartesian :-** From $A(r, \theta, \phi) \rightarrow A(x, y, z)$

In this a vector given in spherical co-ordinate is to be converted in cartesian system. i.e.

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Ist r, θ & ϕ must be converted in terms of x, y & z. For that x, y & z must be replaced by.

$$\boxed{r = \sqrt{x^2+y^2+z^2},\quad \theta = \cos^{-1}\frac{z}{r} = \cos^{-1}\frac{z}{\sqrt{x^2+y^2+z^2}} \qquad \phi = \tan^{-1}\left(\frac{y}{x}\right)}$$

$$\sin\theta = \frac{\rho}{r} = \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \qquad \sin\phi = \frac{y}{\rho} = \frac{y}{\sqrt{x^2+y^2}}$$

$$\cos\theta = \frac{z}{r} \Rightarrow \frac{z}{\sqrt{x^2+y^2+z^2}} \qquad \cos\phi = \frac{x}{\rho} = \frac{x}{\sqrt{x^2+y^2}}$$

(3) Cartesian to Cylinderical :- From $A(x,y,z) \rightarrow A(\rho,\phi,z)$

In this,

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Also, x, y & z must be replaced I^{st} by,

$$\boxed{x = \rho\cos\phi \;,\; y = \rho\sin\phi \;\&\; z = z}$$

(4) Cylinderical to Cartesian :- From $\vec{A}(\rho,\phi,z) \rightarrow \vec{A}(x,y,z)$

In this,

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

Also ρ, ϕ & z must be replaced I^{st} by,

$$\boxed{\rho = \sqrt{x^2+y^2} \;,\; \phi = \tan^{-1}\left(\frac{y}{x}\right) \;\&\; z = z}$$

$$\boxed{\sin\phi = \frac{y}{\rho} \qquad \cos\phi = \frac{x}{\rho}}$$

(5) Cylinderical to Spherical :- From $A(\rho,\phi,z) \rightarrow A(r,\theta,\phi)$

$$\Rightarrow \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

and ρ, ϕ & z must be replaced I^{st} by,

$$\boxed{\rho = r\sin\theta \;,\; \phi = \phi \;\text{ and }\; z = r\cos\theta}$$

(6) Spherical to Cylinderical :- From $A(r,\theta,\phi) \rightarrow A(\rho,\phi,z)$

$$\Rightarrow \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

& r, θ & ϕ must be replaced I^{st} by,

$$\boxed{r = \sqrt{\rho^2+z^2} \;,\; \theta = \cos^{-1}\left(\frac{z}{\sqrt{\rho^2+z^2}}\right) \;,\; \phi = \phi.}$$

$$\sin\theta = \frac{\rho}{r} = \frac{\rho}{\sqrt{\rho^2+z^2}} \qquad \cos\theta = \frac{z}{r} = \frac{z}{\sqrt{\rho^2+z^2}}$$

LINE INTEGRAL:- If any quantity is distributed over a line then total quantity can be obtained by taking line integral.

It is the integral over a line (Linear Path).

$$W = \int_P^Q F \cdot dl$$ This integral is performed along the path P to Q

For a closed path, $W = \oint F \cdot dl$

$dl = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ (In Cartesian co-ordinate System)

$dl = d\rho\,\hat{\rho} + \rho\, d\phi\,\hat{\phi} + dz\,\hat{z}$ (In Cylinderical co-ord. System)

$dl = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\, d\phi\,\hat{\phi}$ (In spherical system)

SURFACE INTEGRAL:-

It is integral over a surface gives total quantity if it is distributed uniformly over a surface.

$$I = \int_S J \cdot dS = \iint J \cdot dS$$

For cartesian system:-

$$dS = dS_x\,\hat{x} + dS_y\,\hat{y} + dS_z\,\hat{z}$$

where $dS_x = dy\,dz$, $dS_y = dx\,dz$ and $dS_z = dx\,dy$

For Cylinderical System:-

$$dS = dS_\rho\,\hat{\rho} + dS_\phi\,\hat{\phi} + dS_z\,\hat{z}$$

$dS_\rho = \rho\, d\phi\, dz$, $dS_\phi = d\rho\, dz$, $dS_z = d\rho\, \rho\, d\phi = \rho\, d\rho\, d\phi$

For Spherical System:-

$$dS = dS_r\,\hat{r} + dS_\theta\,\hat{\theta} + dS_\phi\,\hat{\phi}$$

$dS_r = r\,d\theta\; r\sin\theta\, d\phi = r^2 \sin\theta\, d\theta\, d\phi$

$dS_\theta = dr\; r\sin\theta\, d\phi = r\sin\theta\, dr\, d\phi$

$dS_\phi = dr\; r\,d\theta = r\, dr\, d\theta$

VOLUME INTEGRAL:- If a quantity is distributed over a volume then total quantity can be obtained by taking volume integral.

$$m = \int_V \rho\, dV = \iiint \rho\, dV$$

where $dV = dx\,dy\,dz$ (In Cartesian System)

$dV = d\rho\; \rho\, d\phi\, dz = \rho\, d\phi\, d\rho\, dz$ (In Cylinderical System)

$dV = dr\; r\,d\theta\; r\sin\theta\, d\phi = r^2 \sin\theta\, dr\, d\theta\, d\phi$ (In Spherical System).

DEL OPERATOR :-

It is a vector space function operator and is defined through the spatial derivatives w.r.t. the space co-ordinates. In cartesian co-ordinates,

$$\nabla = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$$

Product of this ∇ with a scalar field gives

$$\text{Grad}\,\phi = \nabla\phi = \frac{\partial\phi}{\partial x}\hat{x} + \frac{\partial\phi}{\partial y}\hat{y} + \frac{\partial\phi}{\partial z}\hat{z}$$

This is known as Gradient of scalar function ϕ.

Product of ∇ with vector gives two terms as

Dot product of ∇ with a vector gives divergence of that vector field. i.e.

$$\text{Div.}\,\vec{A} = \nabla\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Cross product of ∇ with a vector field gives curl of that vector field. i.e.

$$\text{Curl}\,\vec{A} = \nabla\times\vec{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$$

Laplacian of a scalar field is defined as the ~~gra~~ divergence of the gradient of a scalar i.e.

$$\nabla\cdot\nabla\phi = \nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is known as Laplacian operator.

GRADIENT, DIV. & CURL AND LAPLACIAN IN DIFFERENT CO-ORDINATE SYSTEMS :- (9)

To derive expression for Grad; Div. & Curl and Laplacian operator can be obtained using some general variables which are given below:-

	General SYSTEM	Cartesian co-ordinate	Cylindrical co-ordinate	Spherical co-ordinate
Variables	u_1 u_2 u_3	x y z	ρ ϕ z	r θ ϕ
Scale Factors	h_1 h_2 h_3	1 1 1	1 ρ 1	1 r $r\sin\theta$
Unit Vectors	$\hat{e}_1$ $\hat{e}_2$ $\hat{e}_3$	$\hat{x}$ $\hat{y}$ $\hat{z}$	$\hat{\rho}$ $\hat{\phi}$ $\hat{z}$	$\hat{r}$ $\hat{\theta}$ $\hat{\phi}$
Vector component	A_1 A_2 A_3	A_x A_y A_z	A_ρ A_ϕ A_z	A_r A_θ A_ϕ

If $\boxed{dl = dl_1\,\hat{e}_1 + dl_2\,\hat{e}_2 + dl_3\,\hat{e}_3}$

then $\boxed{dl_1 = h_1\,du_1 \;,\; dl_2 = h_2\,du_2 \;,\; dl_3 = h_3\,du_3}$

and $\boxed{dv = dl_1\,dl_2\,dl_3 = h_1 h_2 h_3\,du_1\,du_2\,du_3}$

$$dS = dS_1\,\hat{e}_1 + dS_2\,\hat{e}_2 + dS_3\,\hat{e}_3$$

where, $dS_1 = h_2\,du_2\,h_3\,du_3 = h_2 h_3\,du_2\,du_3$

$dS_2 = h_1\,du_1\,h_3\,du_3 = h_1 h_3\,du_1\,du_3$

$dS_3 = h_1\,du_1\,h_2\,du_2 = h_1 h_2\,du_1\,du_2$

Gradient :- It can be given in general system as

$$\boxed{\nabla\phi = \hat{e}_1 \frac{1}{h_1}\frac{\partial \phi}{\partial u_1} + \hat{e}_2 \frac{1}{h_2}\frac{\partial \phi}{\partial u_2} + \hat{e}_3 \frac{1}{h_3}\frac{\partial \phi}{\partial u_3}}$$

For example, in Cartesian System,

$\hat{e}_1, \hat{e}_2, \hat{e}_3 = \hat{x}, \hat{y} \,\&\, \hat{z}$ and $h_1, h_2, h_3 = 1, 1, 1,$

and $u_1, u_2 \,\&\, u_3 = x, y \,\&\, z$

$$\Rightarrow \nabla\phi = \hat{x}\frac{1}{1}\frac{\partial \phi}{\partial x} + \hat{y}\frac{1}{1}\frac{\partial \phi}{\partial y} + \hat{z}\frac{1}{1}\frac{\partial \phi}{\partial z}$$

$$\Rightarrow \nabla\phi = \frac{\partial \phi}{\partial x}\hat{x} + \frac{\partial \phi}{\partial y}\hat{y} + \frac{\partial \phi}{\partial z}\hat{z}$$

Divergence :-

$$\boxed{\nabla\cdot\vec{A} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(A_1h_2h_3) + \frac{\partial}{\partial u_2}(A_2h_1h_3) + \frac{\partial}{\partial u_3}(A_3h_1h_2)\right]}$$

e.g. in cylinderical system.

$h_1, h_2, h_3 = 1, \rho, 1$ and $u_1, u_2, u_3 = \rho, \phi, z$ and $\hat{e}_1, \hat{e}_2, \hat{e}_3 = \hat{\rho}, \hat{\phi}, \hat{z}$

$A_1, A_2, A_3 = A_\rho, A_\phi, A_z$

$$\Rightarrow \nabla\cdot\vec{A} = \frac{1}{1\times\rho\times1}\left[\frac{\partial}{\partial\rho}(A_\rho\,\rho) + \frac{\partial}{\partial\phi}(A_\phi) + \frac{\partial}{\partial z}(A_z\,\rho)\right]$$

$$\nabla\cdot\vec{A} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi} + \frac{\cancel{\rho}}{\cancel{\rho}}\frac{\partial A_z}{\partial z}$$

Curl :-

$$\nabla\times\vec{A} = \frac{1}{h_1h_2h_3}\begin{vmatrix} \hat{e}_1 h_1 & \hat{e}_2 h_2 & \hat{e}_3 h_3 \\ \partial/\partial u_1 & \partial/\partial u_2 & \partial/\partial u_3 \\ A_1 h_1 & A_2 h_2 & A_3 h_3 \end{vmatrix}$$

e.g. in spherical system,

$$\nabla\times\vec{A} = \frac{1}{1\times r\times r\sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\,\hat{\phi} \\ \partial/\partial r & \partial/\partial\theta & \partial/\partial\phi \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix} = \begin{vmatrix} \frac{\hat{r}}{r^2\sin\theta} & \frac{r\hat{\theta}}{r^2\sin\theta} & \frac{r\sin\theta\,\hat{\phi}}{r^2\sin\theta} \\ \partial/\partial r & \partial/\partial\theta & \partial/\partial\phi \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

$$\nabla\times\vec{A} = \begin{vmatrix} \frac{\hat{r}}{r^2\sin\theta} & \frac{\hat{\theta}}{r\sin\theta} & \frac{\hat{\phi}}{r} \\ \partial/\partial r & \partial/\partial\theta & \partial/\partial\phi \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

Laplacian :-

$$\boxed{\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial u_3}\right)\right]}$$

e.g. in cylindrical system, $h_1, h_2, h_3 = 1, \rho, 1$

$$\nabla^2 f = \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}\left(\rho \frac{\partial f}{\partial \rho}\right) + \frac{\partial}{\partial \phi}\left(\frac{1}{\rho}\frac{\partial f}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(\rho \frac{\partial f}{\partial z}\right)\right]$$

$$\boxed{\nabla^2 f = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\cancel{\rho}}{\cancel{\rho}}\frac{\partial^2 f}{\partial z^2}}$$

PHYSICAL INTERPRETAION (SIGNIFICANCE) OF GRADIENT, DIVERGENCE & CURL :-

GRADIENT :-

It is the maximum space rate of change of the scalar function. If V is potential then ∇V is Potential gradient and gives rate of change of potential with distance i.e. change in potential per unit length.

As gradient of a scalar function will give a vector so it also have the direction. The direction of the ∇V is the direction in which potential (Scalar function) is changing most rapidly.

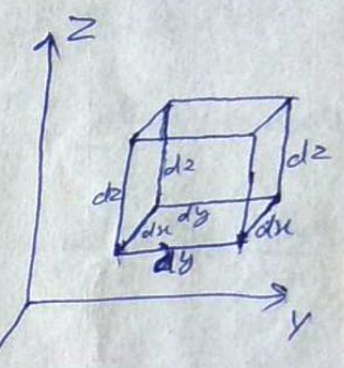
$$\nabla V = \frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}$$

For potential, unit of ∇V is volt/meter.

DIVERGENCE :-

Divergence is the dot product of ∇ and some vector field. As dot product results in a scalar, so $\nabla \cdot \vec{A}$ ($\vec{A}$ is some arbitrary vector field) don't have any direction.

For interpretation of divergence, consider the flow of some fluid. Consider a small elemental volume $dv = dx\,dy\,dz$ with in the fluid as shown in fig.



If ρ_m is the mass density of the fluid, the flow into the volume through left hand face is $\rho_m v_y\,dx\,dz$.

where v_y is the y-compo of velocity, v, of the fluid.

$\left[\because \rho_m = \frac{\text{mass of fluid}}{\text{Volume}} \right.$ Total ~~value~~ mass entered $= \rho_m \times$ Volume

If the gradient of velocity in y-direction is $\frac{\partial v_y}{\partial y}$, then ~~velo~~ velocity at right hand face can be given as

$$v_y' = \cancel{\rho_m}\, v_y + \frac{\partial v_y}{\partial y} dy$$

$\frac{\partial v_y}{\partial y} dy$ -> change in velocity in dy distance.

Now, fluid leaving the volume from right hand face can be given by $\rho_m v_y'\,dx\,dz$.

So, Net outward flow in y-direction is,

$$\rho_m v_y' dx\,dz - \rho_m v_y\,dx\,dz$$

$$= \rho_m \left[v_y + \frac{\partial v_y}{\partial y} dy \right] dx\,dz - \rho_m v_y\,dx\,dz$$

$$= \left[\rho_m v_y + \rho_m \frac{\partial v_y}{\partial y} dy \right] dx\,dz - \rho_m v_y\,dx\,dz$$

$$= \cancel{\rho_m v_y\,dx\,dz} + \frac{\partial (\rho_m v_y)}{\partial y} dx\,dy\,dz - \cancel{\rho_m v_y\,dx\,dz}$$

$\left[\because \rho_m \frac{\partial v_y}{\partial y} = \frac{\partial (\rho_m v_y)}{\partial y} \right.$ As ρ_m = constant

$$= \frac{\partial (\rho_m v_y)}{\partial y} dx\,dy\,dz$$

Similarly, the net outward flow in z direction is

$$\frac{\partial(\rho_m V_z)}{\partial z}\, dx\,dy\,dz$$

Also, net outward flow in x direction is

$$\frac{\partial(\rho_m V_x)}{\partial x}\, dx\,dy\,dz$$

So, The total outward flow ~~is~~ must be sum of these three.

i.e. Total outward flow $= \left[\frac{\partial(\rho_m V_x)}{\partial x} + \frac{\partial(\rho_m V_y)}{\partial y} + \frac{\partial(\rho_m V_z)}{\partial z}\right] dx\,dy\,dz$

So, the Net outward flow/volume $= \frac{\partial(\rho_m V_x)}{\partial x} + \frac{\partial(\rho_m V_y)}{\partial y} + \frac{\partial(\rho_m V_z)}{\partial z}$

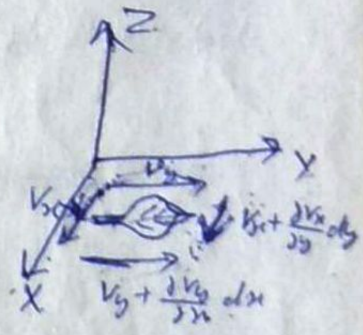
$$= \nabla\cdot(\rho_m \vec{V})$$

Hence, Divergence ~~of~~ gives the net outward flow/volume.

If $\nabla\cdot\vec{A} = 0$ then vector field $\vec{A}$ is said to be <u>Solenoidal</u> or <u>Incompressible</u>.

CURL :-

The concept of curl or rotation is illustrated in the stream flow problems.



Consider a stream on the surface of which a leaf is floating in x-y plane as shown in fig. If the velocity is only in y-direction then there will be only translational & no rotational motion. But if there are eddies in the stream flow, there will be a rotational ~~also~~ as well as translational motion will be there. The rotation about z-direction can be given by $(\nabla\times V)_z$ i.e. z-component of curl $\vec{V}$.

From fig. it is clear that a +ve value of $\frac{\partial V_y}{\partial x}$ will tend to rotate leaf in anti clock wise direction and a +ve value of $\frac{\partial v_x}{\partial y}$ in clockwise direction. So, net amount of rotation will be proportional to the difference of these two if initially no rotation is assumed. The rotation from x-axis to y-axis is always taken +ve rotation, hence amount of rotation about z-axis can be given as,

$$(\nabla \times V)_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}$$

Similarly, Rotation about x-axis can be given as,

$$(\nabla \times V)_x = \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}$$

Also, Rotation about y-axis is,

$$(\nabla \times V)_y = \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}$$

since, the ~~directions~~ rotation have magnitude as well as direction, thus, total amount of rotation about any axis will be the vector ~~sum~~ of the rotation about x, y & z axis.

i.e. Net Rate of Rotation or Angular velocity is proportional to,

$$\nabla \times V = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{x} + \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right)\hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\hat{z}$$

The direction of the resultant curl will be the axis of rotation.

If $\nabla \times \vec{A} = 0$, it means field is irrotational.

$\vec{A} \rightarrow$ Arbitrary vector field.

UNIT - 1/CH-2-ELECTROSTATICS

Point charge :- These are very small charges, assumed to be of infinitesimally small volume.

Charge Distribution :- There are 3 types of charge distribution-

(I) **Line charge distribution** :- It is described in terms of line charge density (ρ_l) C/m.

ρ_l is defined as:-

$$\rho_l = \lim_{\Delta l \to 0} \frac{\Delta Q}{\Delta l} \ C/m$$

where ΔQ = small amount of charge
Δl = small length.

or $$Q = \int_l \rho_l \, dl.$$

(II) **Surface Charge Density/Distribution** :- It is visualised as charge distributed on a conducting sheet, like charge distributed on plates of capacitor. Surface charge Density (ρ_s) C/m² can be given as

$$\rho_s = \lim_{\Delta S \to 0} \frac{\Delta Q}{\Delta S} \ C/m^2, \quad \Delta S = \text{Small surface area.}$$

or $$Q = \int_S \rho_s \, dS.$$

(III) **Volume Charge Distribution** :- This can be seen as a region of space filled with charge. The volume charge density (ρ_v) C/m³ can be given as:-

$$Q = \int_v \rho_v \, dV$$

or $$\rho_v = \lim_{\Delta V \to 0} \frac{\Delta Q}{\Delta V}$$

where ΔV = Small Elementary volume

2

Coulomb's Law :- This law gives the force between two charges separated by some finite distance in a given medium. It states that,

$$F \propto Q_1 Q_2$$

$$F \propto \frac{1}{r^2}$$

where Q_1 & Q_2 = charges
r = Distance between charges.

$$\Rightarrow F \propto \frac{Q_1 Q_2}{r^2}$$

$$\Rightarrow F = K \frac{Q_1 Q_2}{r^2}$$

For SI system of units -

$$K = \frac{1}{4\pi\varepsilon}$$

where ε = Electric Permittivity of medium
= Dielectric constant of medium

$$\Rightarrow F = \frac{Q_1 Q_2}{4\pi\varepsilon r^2}$$

Also $\varepsilon = \varepsilon_0 \varepsilon_r$; ε_0 = Dielectric constant of free space

$$\Rightarrow F = \frac{Q_1 Q_2}{4\pi\varepsilon_0\varepsilon_r r^2} \text{ N}$$

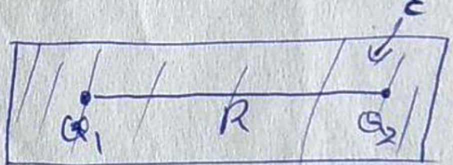
$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

$\varepsilon_0 = \frac{1}{36\pi \times 10^9}$ F/m

ε_r = Relative Permittivity of Medium

In vector form,

$$\vec{F_2} = \frac{Q_1 Q_2}{4\pi\varepsilon R_{12}^2} \hat{R_2}$$



Here $\hat{r}_{12} = \frac{\vec{R_{12}}}{|\vec{R_{12}}|} = \frac{\vec{r_2} - \vec{r_1}}{|\vec{r_2} - \vec{r_1}|}$

③

Electric Field Intensity :- ($\vec{E}$) :-

Consider a +ve charge Q_1 placed at origin. and another +ve charge q known as test charge is brought near Q_1. The Force on q will be,

$$\vec{F} = \frac{Q_1 q}{4\pi\epsilon_0 R^2}\hat{R}$$, where R= Distance b/w Q_1 & q.

It can be said that Q_1 has a field around it which exerts force on q. This field is known as ELECTRIC FIELD due to Q_1. This can be given Mathematically,

$$\vec{E} = \frac{\vec{F}}{q} = \frac{Q}{4\pi\epsilon_0 R^2}\hat{R}$$

$$\vec{E} = \frac{Q \times \vec{R}}{4\pi\epsilon R^3} \qquad \left[\because \hat{R} = \frac{\vec{R}}{|\vec{R}|} = \frac{\vec{R}}{R}\right]$$

If r' & r are position vector of Q_1 & q respectivly, then

$$\vec{R} = \vec{r} - \vec{r}' = (x-x')\hat{x} + (y-y')\hat{y} + (z-z')\hat{z}$$

$$\Rightarrow \vec{E} = \frac{Q(\vec{r}-\vec{r}')}{4\pi\epsilon|\vec{r}-\vec{r}'|^3}$$

Here $|\vec{r}-\vec{r}'| = \sqrt{(x-x')^2+(y-y')^2+(z-z')^2}$

Electric (Scalar) Potential :- It is defined as the work done on the unit +ve charge in moving it from infinity to the point at which potential is to be determined.

The unit of potential is volt [1volt = 1 J/C]

If a small test charge q is to bring in the field of a +ve charge Q from infinity to a distance R from Q, then work done is

$$W = -\int_{\infty}^{R} F \cdot dr$$

$$= -\int_{\infty}^{R} \frac{Qq}{4\pi\epsilon r^2}\, dr$$

$$= -\frac{Qq}{4\pi\epsilon}\left[-\frac{1}{r}\right]_{\infty}^{R}$$

$$= \frac{Qq}{4\pi\epsilon R}$$

As potential is work done per unit charge, hence,

$$\boxed{V = \frac{W}{q} = \frac{Q}{4\pi\epsilon R}}$$

where V = Potential due to Q at a distance R

$\vec{E}$ as Gradient of V. ($E = -\nabla V$) :- consider two points separated by small distance dl. Then work done in moving an unit +ve charge from one point to another is

$$dw = dV = -E \cdot dl \qquad \left[\because dV = \frac{dw}{dq} = -\frac{dF \cdot dl}{dq} = -E\,dl\right]$$

$$\Rightarrow \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz = -E \cdot dl$$

$$\Rightarrow \left(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}\right) \cdot (dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}) = -E \cdot dl$$

$$\Rightarrow \nabla V \cdot dl = -E \cdot dl \Rightarrow \boxed{E = -\nabla V = -\text{Grad } V}$$

Electric Potential Difference :- Consider two points A & B at a radial distance of r_A & r_B from a point charge Q, then

$$V_{BA} = -\int_B^A \vec{E}\cdot d\vec{r} = -\int_{r_B}^{r_A} \frac{Q}{4\pi\epsilon r^2}\, dr$$ [∵ Both $\vec{E}$ & $d\vec{r}$ are in same direction]

$$\Rightarrow V_{BA} = \frac{Q}{4\pi\epsilon}\left[\frac{1}{r}\right]_{r_B}^{r_A}$$ [$\because \int \frac{1}{r^2}dr = -\frac{1}{r}$]

$$\Rightarrow V_{BA} = \frac{Q}{4\pi\epsilon}\left[\frac{1}{r_A} - \frac{1}{r_B}\right]$$

$$\boxed{\begin{aligned} V_{BA} &= \frac{Q}{4\pi\epsilon r_A} - \frac{Q}{4\pi\epsilon r_B} \\ V_{BA} &= V_A - V_B \end{aligned}}$$

$\vec{E}$ & V Due to Several Charges :- Let us consider a system of n charges $Q_1, Q_2, Q_3 - - Q_n$ which are situated at a distance of $r_1, r_2, - - r_n$ respectively from point P at which $\vec{E}$ & V is to be determined. Then, $\vec{E}$ & V can be given as the sum of $\vec{E}$ & V due to individual charges like

$$\vec{E} = \frac{Q_1}{4\pi\epsilon r_1^2}\hat{r}_1 + \frac{Q_2}{4\pi\epsilon r_2^2}\hat{r}_2 + - - - + \frac{Q_n}{4\pi\epsilon r_n^2}\hat{r}_n$$

$$\boxed{\vec{E} = \frac{1}{4\pi\epsilon}\sum_{i=1}^{n}\frac{Q_i}{r_i^2}\hat{r}_i}$$

[Remember the sum is vector sum of $E_1, E_2 - - E_n$.]

IIIly, $$V = \frac{Q_1}{4\pi\epsilon r_1} + \frac{Q_2}{4\pi\epsilon r_2} + - - + \frac{Q_n}{4\pi\epsilon r_n}$$

$$\Rightarrow V = \frac{1}{4\pi\epsilon}\sum_{i=1}^{n}\frac{Q_i}{r_i}$$ [This is algebric sum as V is a scalar quantity]

(6)

If charge is distributed continuously on a line and line charge density is ρ_ℓ C/m, then

$$V_L = \frac{1}{4\pi\varepsilon}\int_\ell \frac{\rho_\ell}{r}\,d\ell \text{ and } E_L = \frac{1}{4\pi\varepsilon}\int_\ell \frac{\rho_\ell}{r^2}\,d\ell\,\hat{r}$$

||ly for surface & volume charge distribution

$$V_s = \frac{1}{4\pi\varepsilon}\int_S \frac{\rho_s}{r}\,dS \;\&\; V_v = \frac{1}{4\pi\varepsilon}\int_v \frac{\rho_v}{r}\,dv$$

Electric Flux Density :- ($\vec{D}$) And Electric Flux (ψ) :-

It is seen that $\vec{E}$ depends not only on the charge Q but also depends upon medium in which charge is placed. So, It is desirable to associate with charge Q a second electrical quantity that is independent of the medium. This quantity is known as **Electric Displacement** or **Electric Flux** (ψ)

This can be explained by Faraday's experiment of two spheres. A sphere with charge Q is placed within, but not touching, a larger hollow sphere. The outer sphere is 'earthed' momentarily and then inner sphere is removed and charge on outer sphere is then measured. This is found to be equal (But of opposite sign) to the charge on inner sphere for all sizes of spheres and for all types of dielectric media b/w the spheres. Thus, it could be considered that there is an Electric Displacement from charge on inner sphere to the outer sphere through the medium and amount of displacement depends only upon magnitude of charge Q.

(7)

Electric Displacement Density ($\vec{D}$) ~~de~~ at any point on a spherical surface of radius r, having a charge Q at the center can be given as,

$$\vec{D} = \frac{\psi}{4\pi r^2} = \frac{Q}{4\pi r^2} = \epsilon \vec{E}$$

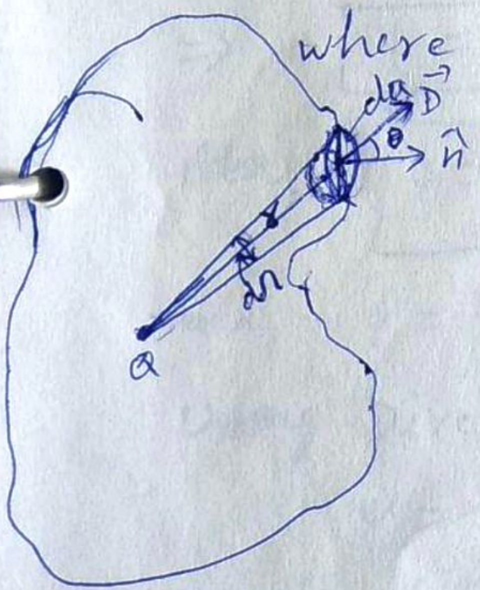
Hence, $\vec{D}$ is a vector quantity having direction same as that of $\vec{E}$.

Gauss's Law :- It states that the total displacement or Electric flux through a closed surface is equal to the amount of charge enclosed by that surface.

Proof :- Consider the displacement through the small element da of surface S is,

$$d\psi = D \cdot da = D\, da \cos\theta \quad \text{— (1)}$$

where θ = angle b/w $\vec{D}$ & normal to da (Area vector)



From fig. it is seen that da cosθ is the projection of da, normal to the radius vector.

Thus, By the definition of solid angle,

$$\frac{da\cos\theta}{r^2} = d\Omega = \text{Solid angle subtended at Q by area da.}$$

$$\Rightarrow da\cos\theta = r^2 d\Omega \quad \text{— (2)}$$

Now, Total displacement flux through surface S is obtained ~~as~~ from eq. (1) as,

$$\psi = \oint_S D\, da \cos\theta \quad \text{—— (3)}$$

From eq. 2, eq. 3 can be given as,

$$\psi = \oint_S D r^2 d\Omega$$

we know ~~that~~ By the definition of $\vec{D}$ that

$$D = \frac{Q}{4\pi r^2}$$

$$\Rightarrow \psi = \oint_S \frac{Q}{4\pi r^2} \cdot r^2 d\Omega$$

$$\Rightarrow \psi = \frac{Q}{4\pi} \oint_S d\Omega$$

The total solid angle subtended by a closed surface is,

$$\Omega = \oint_S d\Omega = 4\pi \text{ steradians.}$$

$$\Rightarrow \psi = \frac{Q}{4\pi} \times 4\pi$$

$$\Rightarrow \boxed{\psi = Q.}$$

Also $\boxed{\psi = \int_V \rho_v dv}$ $\left[\because Q = \int_V \rho dv\right]$

and $\psi = \int_S D \cdot da$

Using Divergence theorem,

$$\psi = \oint_S D \cdot da = \int_V (\nabla \cdot D) dv$$

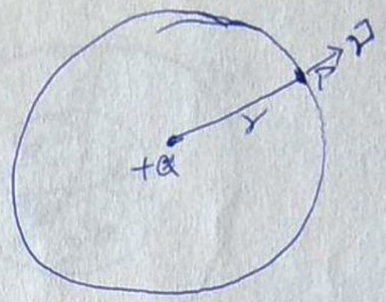
$$\Rightarrow \int_V (\nabla \cdot D) dv = \int_V \rho dv$$

$$\Rightarrow \boxed{\nabla \cdot D = \rho}$$ → This is the differential form of Gauss's law.

APPLICATIONS OF GAUSS's LAW :-

(I) Field Due to a point charge :-

Let us consider a point charge Q. The field will be spherical and radially outward. The Total flux outwards through the surface is (By Gauss's Law),



$$\int d\psi = \oint_s \vec{D}\cdot d\vec{s} = \int_v \rho_v \, dv = Q$$

$$\psi = \oint_s \varepsilon_0 \vec{E}\cdot d\vec{s} = Q$$

$$\Rightarrow \oint_s \varepsilon_0 (\vec{E_r}\,\hat{r})\cdot(ds\,\hat{r}) = Q$$

$$\Rightarrow \oint_s \varepsilon_0 E_r \, ds = Q \qquad [\because \hat{r}\cdot\hat{r} = 1]$$

$$\Rightarrow \varepsilon_0 E_r \oint_s ds = Q$$

$$\Rightarrow \varepsilon_0 E_r \cdot 4\pi r^2 = Q$$

$$\Rightarrow E_r = \frac{Q}{4\pi\varepsilon_0 r^2}$$

Also,

$$\vec{E} = E_r \,\hat{r}$$

$$\Rightarrow \boxed{\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\,\hat{r}} \quad V/m \text{ or } N/C$$

$$V = -\int_\infty^r \vec{E}\cdot d\vec{r} = -\int_\infty^r E_r \, dr$$

$$= -\int_\infty^r \frac{Q}{4\pi\varepsilon_0 r^2} dr = -\frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_\infty^r$$

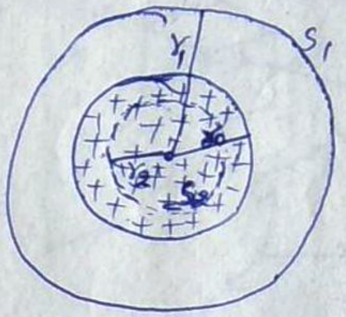
$$\boxed{V = \frac{Q}{4\pi\varepsilon_0 r}}$$

II Field of a charge cloud :-

For a spherical cloud having a uniform charge density ρ_v C/m³, two cases arises -

(a) The field outside the cloud i.e. $r > r_0$:-

In this case the charge enclosed by surface S_1 of radius r_1 will be $Q = \int_v \rho_v dv$



$$\Rightarrow \oint_{S_1} \vec{D} \cdot \vec{ds} = \int_v \rho_v dv$$

$$\Rightarrow \varepsilon_0 E_r \cdot 4\pi r_1^2 = \rho_v \left(\frac{4}{3} \pi r_0^3\right)$$

$$\Rightarrow E_r = \frac{\rho_v \left(\frac{4}{3} \pi r_0^3\right)}{\varepsilon_0 \, 4\pi r_1^2}$$

$$\Rightarrow E_r = \frac{\rho_v r_0^3}{3 \varepsilon_0 r_1^2}$$

$$\Rightarrow \boxed{\vec{E} = \frac{\rho_v r_0^3}{3 \varepsilon_0 r_1^2} \hat{r}}$$

(b) Field inside the cloud i.e. $r < r_0$:-

Applying Gauss's law to closed surface S_2 of radius r_2, we get

$$\oint_{S_2} \vec{D} \cdot \vec{ds} = \int_v \rho_v dv$$

$$\Rightarrow \varepsilon_0 E_r \, 4\pi r_2^2 = \rho_v \int_v dv = \rho_v \left[\frac{4}{3} \pi r_2^3\right]$$

$$\Rightarrow E_r = \frac{\rho_v \left[\frac{4}{3} \pi r_2^3\right]}{\varepsilon_0 \, 4\pi r_2^2}$$

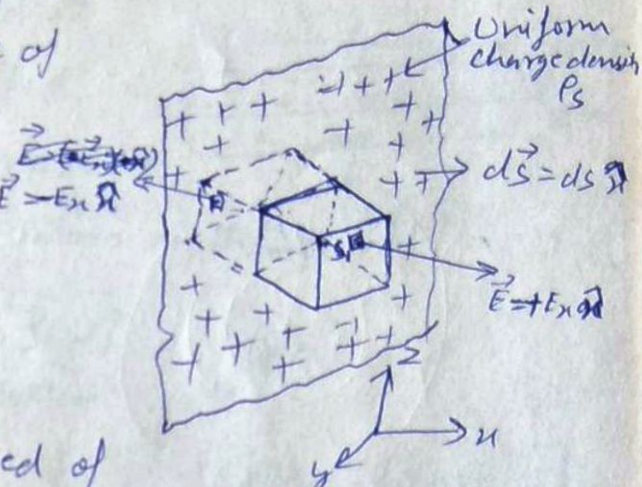
$$\Rightarrow E_r = \frac{\rho_v r_2}{3 \varepsilon_0} \Rightarrow \boxed{\vec{E} = \frac{\rho_v r_2}{3 \varepsilon_0} \hat{r}} \text{ V/m}$$

(c) At the surface $r = r_0$:-

$$\boxed{\vec{E} = \frac{\rho_v r_0}{3 \varepsilon_0} \hat{r}} \text{ V/m}$$

III Field of an Infinite, Planar charge:-

consider a closed surface S is drawn in the form of a rectangular parallelopiped which is extending equally on either side of Planar charge as shown in fig.



Flux emerges only through S_1 & S_2 ends of parallelopiped as Area vector & $\vec{E}$ are perpendicular on all other surfaces.

If A is the cross-sectional area of the parallelopiped, then on applying Gauss's law,

$$\oint_{S} \rho_S \cdot d\vec{S} = \varepsilon_0\Big[\underset{\text{(Positive side)}}{\oint_{S_1}\vec{E}\cdot d\vec{S}} + \underset{\text{(Negative side)}}{\oint_{S_2}\vec{E}\cdot d\vec{S}}\Big]$$

$$\Rightarrow \varepsilon_0\oint_{S_1}(E_x\hat{x})\cdot(ds\,\hat{x}) + \varepsilon_0\oint_{S_2}(-E_x\hat{x})\cdot(-ds\,\hat{x}) = \oint \rho_S\, ds = \rho_S A$$

$$\Rightarrow \varepsilon_0 E_x\oint_{S_1} ds + \varepsilon_0 E_x\oint_{S_2} ds = \rho_S A$$

$$\Rightarrow \varepsilon_0 E_x A + \varepsilon_0 E_x A = \rho_S A$$

$$\Rightarrow E_x = \frac{\rho_S}{2\varepsilon_0}$$

we know $\vec{E} = E_x\hat{x}$, $x>0$

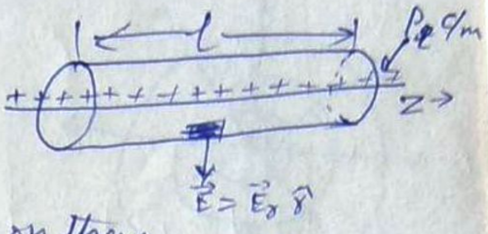
$\vec{E} = -E_x\hat{x}$, $x<0$

$$\Rightarrow \boxed{\vec{E} = +\frac{\rho_S}{2\varepsilon_0}\hat{x} \quad \text{and} \quad \vec{E} = -\frac{\rho_S}{2\varepsilon_0}\hat{x}}$$

(IV) Field of a long line Charge :-

Let us draw a long lin charge of length l. consider a closed cylinder of radius r around it.

Now, $\vec{E}$ will be radially directed and Left side of Gauss's law is zero over end caps of S because $\vec{E}\cdot\vec{ds}$ is zero on them.



$$\Rightarrow \oint_{sides} \vec{D}\cdot\vec{ds} + \oint_{top} \vec{D}\cdot\vec{ds} + \oint_{bottom} \vec{D}\cdot\vec{ds} = \int_l \rho_l \, dl$$

$$\Rightarrow \oint_{sides} \vec{D}\cdot\vec{ds} = \int_l \rho_l \, dl \qquad \left[\because \oint_{top} \vec{D}\, ds = \oint_{Bottom} D\cdot ds = 0\right]$$

$$\Rightarrow \varepsilon_0 \oint_s E_r \, ds = \rho_l \times l \qquad \left[\because \oint_l dl = l\right]$$

$$\Rightarrow \varepsilon_0 \, E_r \, 2\pi r l = \rho_l \times l \qquad \left[\because \oint_s ds = 2\pi r l\right]$$

$$\Rightarrow E_r = \frac{\rho_l}{2\pi\varepsilon_0 r}$$

$$\Rightarrow \boxed{\vec{E} = \frac{\rho_l}{2\pi\varepsilon_0 r}\,\hat{r}}$$

we can show that, $\vec{E}$ does not depend upon length of the line charge, hence we can give $\vec{E}$ due to an infinite long line charge having linear charge density ρ_l as,

$$\boxed{\vec{E} = \frac{\rho_l}{2\pi\varepsilon_0 r}\,\hat{r}}$$

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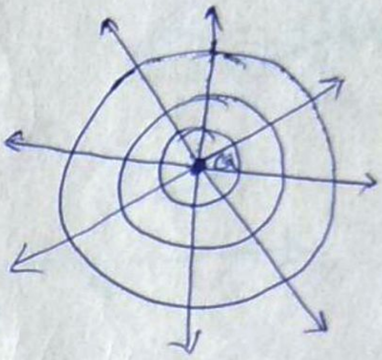
EQUIPOTENTIL SURFACES:- Equipotential surface is that surface at every point of which the potential has the same value.

For example, for a point charge, the equipotential surfaces are spheres with their centre at the charge as shown in fig.



The movement of charge over such a surface requires no work. Any two points on the surface have zero potential difference and hence zero $\vec{E}$ everywhere along (tangential) to surface i.e. the $\vec{E}$ must always be perpendicular to an equipotential surface.

Let us consider a displacement dS on an equipoten. surface, thus work done can be given by

$$\frac{W}{q} = \frac{V(\vec{S}) - V(\vec{S}+d\vec{S})}{q} = 0$$

$$\Rightarrow \quad V(\vec{S}) - V(\vec{S}+d\vec{S}) = 0$$

$$\Rightarrow \quad V(\vec{S}) - \left\{ V(\vec{S}) + \frac{\partial V}{\partial s}\cdot d\vec{S} \right\} = 0$$

$$\Rightarrow \quad -\frac{\partial V}{\partial s}\cdot d\vec{S} = 0 \text{ or } -\nabla V \cdot d\vec{S} = 0$$

$$\Rightarrow \quad \boxed{\vec{E}\cdot d\vec{S} = 0}$$

i.e. $\vec{E}$ must be always perpendicular to the surface.

DIVERGENCE THEOREM :- It relates an integration throughout a volume to an integration over the surface surrounding the volume. This theorem is also known as **Gauss's Theorem**.

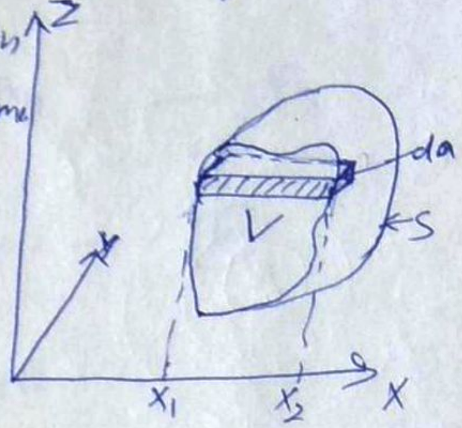


Fig. shows a closed surface S enclosing a volume V that contains charges that produce an electric flux density $\vec{D}$. By definition of divergence, ~~theorem~~

$$\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\int_V (\nabla \cdot \vec{D}) dv = \iiint \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) dx\,dy\,dz \quad -(1)$$

$[\because dv = dx\,dy\,dz]$

Consider now a rectangular volume shown shaded, which has dimensions dy and dz in the y & z directions respectively. Let D_{x_1} & D_{x_2} resp. be the x component of the electric flux entering the left hand side and leaving the right hand side of the volume dv. The total flux emerging is the algebraic difference of these two. But

$$D_{x_2} - D_{x_1} = \int_{x_1}^{x_2} \frac{\partial D_x}{\partial x} dx$$

$$\Rightarrow \iiint \frac{\partial D_x}{\partial x} dx\,dy\,dz = \iint (D_{x_2} - D_{x_1})\, dy\,dz \quad -(2)$$

Now dy dz is x component of surface element da and so above eq. is just the integration of the product of D_x and da_x. By definition of Dot product,

$$\vec{D} \cdot d\vec{a} = D_x da_x + D_y da_y + D_z da_z \quad -(3)$$

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Thus, from eq. (1) & eq. (2), it can be writen as.

$$\iiint \frac{\partial D_y}{\partial y}\, dy\, dx\, dz = \iint D_y\, da_y \quad — (3)$$

and $$\iiint \frac{\partial D_z}{\partial z}\, dz\, dx\, dy = \iint D_z\, da_z \quad — (4)$$

Adding eq. (2), (3) & (4), we get

$$\iiint \frac{\partial D_x}{\partial x}\, dx\,dy\,dz + \iiint \frac{\partial D_y}{\partial y}\, dx\,dy\,dz + \iiint \frac{\partial D_z}{\partial z}\, dx\,dy\,dz$$

$$= \iint D_x da_x + \iint D_y da_y + \iint D_z da_z$$

$$\Rightarrow \iiint \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right) dx\,dy\,dz = \iint (D_x da_x + D_y da_y + D_z da_z)$$

$$\Rightarrow \boxed{\int_v (\nabla \cdot D)\, dv = \int_s \vec{D} \cdot d\vec{a}.}$$

This is the Divergence Theorem.

POISSON'S AND LAPLACE'S EQUATIONS :-

If medium is homogeneous and isotropic have dialectric constant, ε, we can write,

$$\nabla \cdot \vec{D} = \varepsilon \nabla \cdot \vec{E} = \rho$$

$$\Rightarrow \quad \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon}$$

we know that,

$$\vec{E} = -\nabla V$$

$$\Rightarrow \quad -\nabla \cdot \nabla V = \frac{\rho}{\varepsilon}$$

$$\Rightarrow \quad \boxed{\nabla^2 V = -\frac{\rho}{\varepsilon}} \quad \text{—(1)}$$

This is known as Poisson's Equation.

In free space, that is, in a region in which there is no charge ($\rho = 0$), it becomes

$$\boxed{\nabla^2 V = 0} \quad \text{—(2)}$$

This special case for source free region is known as Laplace's Equation.

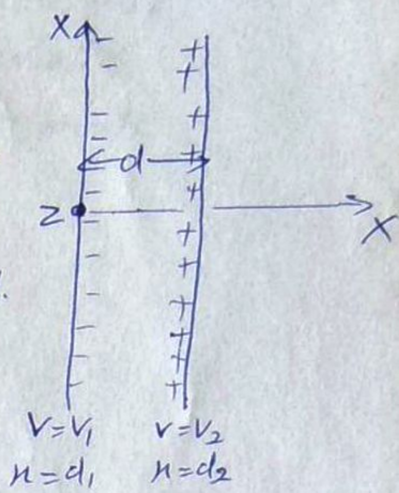
Eq. (2) can also be given as

$$\boxed{\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0}$$

SOLUTION OF LAPLACE'S EQUATION (IN ONE DIMⁿ):-

① Solution in Rectangular co-ordinates:-

Let us consider a parallel plate capacitor. Charge is uniformly distributed over plates i.e. no variation in y & z direction and problem is one dimensional. and Laplace's eq. is



$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

reduces to

$$\frac{\partial^2 V}{\partial x^2} = 0$$

[The partial derivative is also replaced by ordinary derivatives as V is not a function of y & z.]

$$\Rightarrow \frac{d^2 V}{dx^2} = 0 \quad \text{or} \quad \int \frac{d^2 V}{dx^2} = 0$$

$$\Rightarrow \frac{dV}{dx} = A = \text{constant}$$

$$\Rightarrow \int dV = \int A\, dx$$

$$\Rightarrow \boxed{V = Ax + B} \quad \text{—— (1)}$$

where A & B are constants of integration and can be determined as:-

At $x = d_1$ $V = V_1$ and at $x = d_2$ $V = V_2$

$$\Rightarrow V_1 = Ad_1 + B \quad \& \quad V_2 = A d_2 + B$$

$$\Rightarrow V_1 - V_2 = A(d_1 - d_2) \Rightarrow \boxed{A = \frac{V_1 - V_2}{d_1 - d_2}}$$

Also. $V_1 d_2 = A d_1 d_2 + B d_2$ & $V_2 d_1 = A d_2 d_1 + B d_1$

$$\Rightarrow V_1 d_2 - V_2 d_1 = B(d_2 - d_1) \Rightarrow \boxed{B = \frac{V_1 d_2 - V_2 d_1}{d_2 - d_1}}$$

$$\Rightarrow \quad V = \left(\frac{V_1 - V_2}{d_1 - d_2}\right) x + \frac{V_1 d_2 - V_2 d_1}{d_2 - d_1}$$

If we consider boundary condition as,
at $x = 0$ & $V = 0$ on left plate
and $x = d$ & $V = V_0$ on Right plate
then eq. ① will be

$$0 = A(0) + B \Rightarrow B = 0$$

$$\& \quad V_0 = Ad + B \Rightarrow A = \frac{V_0}{d}$$

Hence eq. ① can be written as,

$$\boxed{V = \frac{V_0}{d} x} \text{ Volts}$$

$$\Rightarrow \quad \frac{\partial V}{\partial x} = \frac{V_0}{d}$$

and $E = -\nabla V = -\frac{\partial V}{\partial x}\hat{x}$ $\left[\because \frac{\partial}{\partial y} = \frac{\partial}{\partial z} = 0\right]$

$$\Rightarrow \quad \vec{E} = -\frac{V_0}{d}\hat{x}$$

Also $\vec{D} = \varepsilon\vec{E} \Rightarrow \boxed{\vec{D} = -\varepsilon\frac{V_0}{d}\hat{x}}$

From Gauss law, for a surface charge, we can say

$$D_n = \rho_s \Rightarrow -\frac{\varepsilon V_0}{d} = \rho_s$$

Also $Q = \int \rho_s ds = -\varepsilon\int\frac{V_0}{d}ds = -\frac{\varepsilon V_0 A}{d}$ $\left[\because \int ds = A\right]$

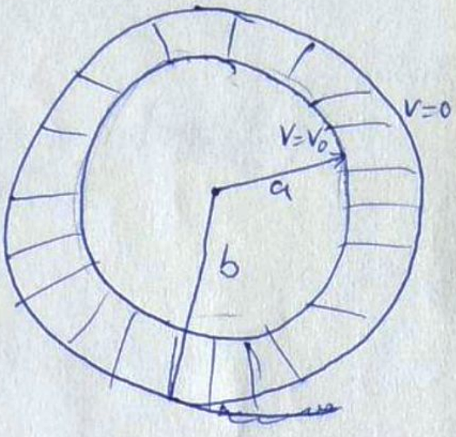
$$|Q| = \left|\frac{-\varepsilon V_0 A}{d}\right| = +\frac{\varepsilon V_0 A}{d}$$

Hence $C = \frac{Q}{V_0} = \frac{\varepsilon A}{d}$ F

This shows the use of Laplace's equation involving minimum mathematics.

② In Cylinderical Co-ordinate System :-

Let us consider a cylinderical capacitor of inner radius 'a' & outer radius 'b' as shown in fig. It is seen that field will vary only w.r.t. ρ and not with ϕ & Z.



Hence, $\nabla^2 V = 0$ in cylinderical coordinates can be writen as,

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) = 0$$

or $$\frac{1}{\rho}\frac{d}{d\rho}\left(\rho \frac{dV}{d\rho}\right) = 0$$

$$\Rightarrow \rho \frac{dV}{d\rho} = \text{Constant} = A \qquad \left[\because \frac{d}{d\rho}(\text{constant}) = 0\right]$$

$$\Rightarrow \frac{dV}{d\rho} = \frac{A}{\rho}$$

$$\Rightarrow \int dV = A\int \frac{d\rho}{\rho}$$

$$\Rightarrow \boxed{V = A \log_e \rho + B} \quad \text{—— ①}$$

Since equipotential surfaces are ρ = constant

$\therefore$ At $\rho = a$, $V = V_0$

& $\rho = b$, $V = 0$

Thus, eq. ① becomes,

$$V_0 = A \log_e a + B$$

$$0 = A \log_e b + B \Rightarrow B = -A \log_e b$$

$$\Rightarrow V_0 = A \log_e a - A \log_e b = A \log_e\left(\frac{a}{b}\right)$$

$$\Rightarrow \boxed{A = \frac{V_0}{\log_e(a/b)}}$$

$\therefore$ eq. 1 becomes,

$$V = \frac{V_0}{\log_e(\frac{a}{b})} \log_e \rho - \frac{V_0}{\log_e(a/b)} \log_e b$$

$$= \frac{V_0}{\log_e(a/b)} \left[\log_e \rho - \log_e b \right] = \frac{V_0}{\log_e(a/b)} \log_e\left(\frac{\rho}{b}\right)$$

$$\Rightarrow \boxed{V = \frac{V_0 \log_e(\rho/b)}{\log_e(a/b)}} \quad \text{--- (2)}$$

But $\vec{E} = -\nabla V$

$$= -\frac{\partial V}{\partial \rho}\hat{\rho} + 0 + 0$$

$$= -\frac{\partial}{\partial \rho}\left[\frac{V_0 \log_e(\rho/b)}{\log_e(a/b)}\right]\hat{\rho} = -\left[\frac{V_0}{\log_e(a/b)} \cdot \frac{1}{\rho/b} \cdot \frac{1}{b}\right]\hat{\rho}$$

$$\Rightarrow \boxed{|\vec{E}| = E = \frac{V_0}{\rho \log_e(b/a)}} \quad \text{--- (3)} \quad \left[\because -\log_e \frac{a}{b} = \log_e b/a\right]$$

Further $\vec{D}_{n\,(\rho=a)} = \varepsilon_0 \vec{E}$

$$= \varepsilon \frac{V_0}{a \log_e(\frac{b}{a})} = \rho_s \qquad [\because \rho = a]$$

$$\Rightarrow Q = \int_s \rho_s \, ds = \int_s \frac{\varepsilon V_0}{a \log_e b/a} ds = \frac{\varepsilon V_0}{a \log_e(b/a)} \int_s ds$$

$$= \frac{\varepsilon V_0}{a \log_e(b/a)} \cdot 2\pi a l \qquad \left[\because \int ds = A = 2\pi a l\right]$$

$$\Rightarrow Q = \frac{2\pi \varepsilon l V_0}{\log_e(b/a)} = \frac{2\pi \varepsilon_0 \varepsilon_r l V_0}{\log_e(b/a)}$$

$$\Rightarrow \boxed{C = \frac{Q}{V_0} = \frac{2\pi \varepsilon_0 \varepsilon_r l}{\log_e(b/a)}}$$

Now, from boundary conditions,

$$V_1 - V_2 = V_{1b} - V_{2b} = 0$$

$$\Rightarrow V_1 - V_2 = 0 \Rightarrow \boxed{V_1 = V_2}$$

Hence, V_1 & V_2 are same and solution is unique.

This theorem can also be applied for ~~Pois~~ Poisson's eq. as $V_{1b} - V_{2b} = 0$ still holds according to boundary conditions. and

$$\nabla^2 V_1 = -\frac{\rho}{\varepsilon} \quad \& \quad \nabla^2 V_2 = -\frac{\rho}{\varepsilon}$$

$$\Rightarrow \nabla^2 (V_1 - V_2) = 0$$

Thus, in the same manner; Poisson's eq. also have unique solution.

CAPACITANCE :-

The capacitance of a capacitor may be defined as "The Ratio of the charge on one of its conductors to the potential difference between them." Symbolically,

$$\boxed{C = \frac{Q}{V}} \quad C/\text{Volt or Farad}$$

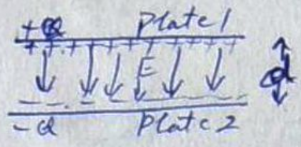
A capacitor is an electric device having two conductors separated by an insulator or dielectric material.

$$C = 1\,F \quad \text{if } Q = 1\,C \ \& \ V = 1\,\text{Volt}.$$

Hence, capacitance of a capacitor is 1 Farad if charge stored is one coulomb with a potential difference of one volt. In practice μF or PicoFarad is used as Farad is a larger capacitance.

Capacitance of Parallel Plate Capacitor:-

Consider a parallel plate capacitor as shown in fig. in which there are two flat parallel plate of area A and distance d b/w them.



+Q = charge on Plate 1, then $\rho_s = \frac{Q}{A}$

Also, $\vec{D}$ in dielectric medium(ϵ) will be equal to ρ_s.

i.e. $D = \rho_s \Rightarrow \epsilon E = \rho_s \Rightarrow E = \rho_s/\epsilon$

where ϵ = dielectric constant of medium b/w plates = $\epsilon_0 \epsilon_r$

Now, Potential difference b/w plates can be given as

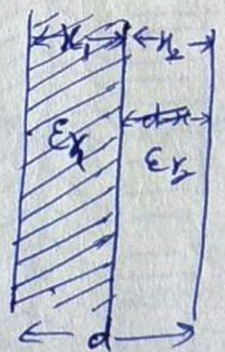
$$V = \int_1^2 E \cdot dl = E \cdot d. \qquad \left[\because E = \frac{V}{d}\right]$$

Now $$C = \frac{Q}{V} = \frac{\rho_s A}{Ed} = \frac{\rho_s A}{\frac{\rho_s}{\epsilon} d} = \frac{\rho_s \epsilon A}{\rho_s d}$$

$$\Rightarrow \boxed{C = \frac{\epsilon A}{d}} \text{ Farad}$$

Capacitance of Parallel Plate Cap.r with two or more dielectric mediums:-

Consider dielectric medium consists of two dielectrics instead of one as shown in fig. Out of these one is air and other is of relative permittivity ϵ_r. Thus $D = \rho_s$ will be same in both media but $\vec{E}$ are different i.e



In medium, $E_1 = \frac{\rho_s}{\epsilon_1} = \frac{\rho_s}{\epsilon_0 \epsilon_{r_1}}$ & $E_2 = \frac{\rho_s}{\epsilon_2} = \frac{\rho_s}{\epsilon_0 \epsilon_{r_2}}$.

where $E_1 = \vec{E}$ of medium of thickness x_1

$E_2 = E$ in medium of thickness x_2.

The Potential difference, V, b/w the plates is,

$$V = E_1 x_1 + E_2 x_2 \qquad [\because V = Ed]$$

$$V = \frac{\rho_s}{\varepsilon_0 \varepsilon_{r_1}} x_1 + \frac{\rho_s}{\varepsilon_0 \varepsilon_{r_2}} x_2 = \frac{\rho_s}{\varepsilon_0}\left[\frac{x_1}{\varepsilon_{r_1}} + \frac{x_2}{\varepsilon_{r_2}}\right] \quad -(1)$$

Also $\rho_s = \frac{Q}{A} \Rightarrow Q = \rho_s A$

$$\therefore \quad V = \frac{Q}{A\varepsilon_0}\left[\frac{x_1}{\varepsilon_{r_1}} + \frac{x_2}{\varepsilon_{r_2}}\right] \quad -(2)$$

Now, $C = \frac{Q}{V} = \dfrac{Q}{\frac{Q}{A\varepsilon_0}\left[\frac{x_1}{\varepsilon_{r_1}} + \frac{x_2}{\varepsilon_{r_2}}\right]}$

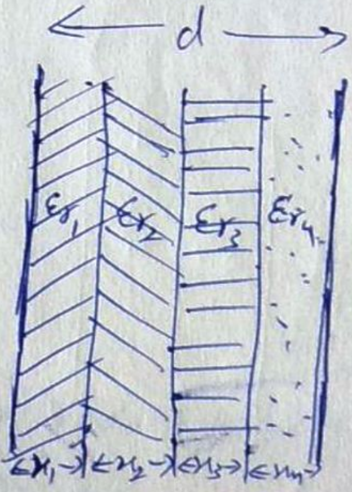
$$\Rightarrow \boxed{C = \frac{\varepsilon_0 A}{\left[\frac{x_1}{\varepsilon_{r_1}} + \frac{x_2}{\varepsilon_{r_2}}\right]}} \quad -(3)$$

Now, Eq (2) can also be written as,

$$\boxed{V = \frac{Q}{A}\left[\frac{x}{\varepsilon_0 \varepsilon_r} + \frac{(d-x)}{\varepsilon_0}\right] = \frac{Q}{A}\left[\frac{x}{\varepsilon} + \frac{(d-x)}{\varepsilon_0}\right]} \quad -(4)$$

Hence, for a capacitor of multiple dielectrics, capacitance can be given ~~using~~ by generalizing eq.(4) as,

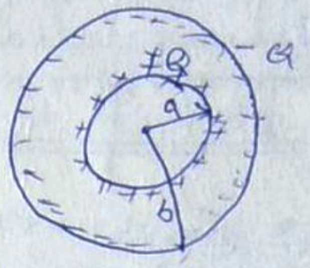
$$\boxed{C = \frac{\varepsilon_0 A}{\frac{x_1}{\varepsilon_{r_1}} + \frac{x_2}{\varepsilon_{r_2}} + \frac{x_3}{\varepsilon_{r_3}} + \cdots} \text{ Farads}}$$



where $\varepsilon_{r_1}, \varepsilon_{r_2}, \ldots$ are relative permittivities of ~~the~~ dielectric materials of thickness $x_1, x_2, x_3, \ldots$ respectively b/w plates of capacitor as shown in fig.

Capacitance of Two Concentric Spheres:

Let us consider a spherical capacitor composed of two concentric sphers of radius a & b of inner & outer shells respectively. Let charge Q is uniformly distributed over the outer surface of inner sphere. and an equal & opposite charge is induced on the outer sphere. as shown in fig.



From Gauss's law,

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{r} = E_r \hat{r}$$

$$\Rightarrow E_r = \frac{Q}{4\pi\epsilon_0 r^2} \qquad a \le r \le b$$

Now,

$$V_{ab} = -\int_b^a E \cdot dr = \int_a^b \frac{Q}{4\pi\epsilon_0 r^2}\,dr$$

$$= \frac{Q}{4\pi\epsilon_0}\left[-\frac{1}{r}\right]_a^b = \frac{Q}{4\pi\epsilon_0}\left[\frac{1}{a}-\frac{1}{b}\right]$$

$$= \frac{Q}{4\pi\epsilon_0}\left[\frac{b-a}{ab}\right]$$

$$C = \frac{Q}{V_{ab}} = \frac{Q}{\frac{Q}{4\pi\epsilon_0}\left(\frac{b-a}{ab}\right)} =$$

$$\boxed{C = \frac{4\pi\epsilon_0 ab}{b-a}}$$

Let A_a, A_b = Area of spherical surface of radius a & b.

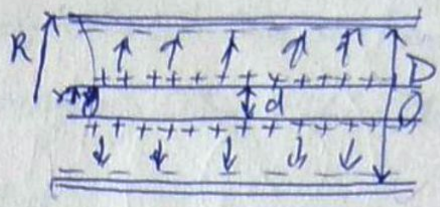
$d = b-a$ = distance b/w them.

$A_a = 4\pi a^2 \Rightarrow a = \sqrt{A_a/4\pi}$ & $A_b = 4\pi b^2 \Rightarrow b = \sqrt{A_b/4\pi}$

$$\Rightarrow C = \frac{4\pi\epsilon_0}{d}\sqrt{\frac{A_a}{4\pi}\cdot\frac{A_b}{4\pi}} = \frac{\epsilon_0}{d}\sqrt{A_a A_b}$$

Capacitance of a Coaxial-Cable:-

For a coaxial cable of long length, ~~the~~ capacitance/length is to be obtained.



Let us assume a dielectric of relative permittivity ϵ_r b/w concentric cylinders of radii R & r respectively.

Let ρ_l is linear charge density on inner & outer conductor.

Hence, $\vec{D}$ can be given as,

$$|\vec{D}| = D = \frac{\rho_l}{2\pi r} \Rightarrow E = \frac{\rho_l}{2\pi\epsilon r}$$

Hence, Potential difference b/w two cylinders is.

$$V = \int_r^R E\,dr = \int_r^R \left(\frac{\rho_l}{2\pi\epsilon r}\right) dr = \frac{\rho_l}{2\pi\epsilon}\left[\log_e r\right]_r^R$$

$$= \frac{\rho_l}{2\pi\epsilon}\log_e\left(\frac{R}{r}\right) = \frac{\rho_l}{2\pi\epsilon}\log_e\left(\frac{D}{d}\right)$$

Now, capacitance/length is,

$$C' = \frac{C}{l} = \frac{Q}{Vl} = \frac{\rho_l}{V}$$

$$= \frac{\rho_l\, 2\pi\epsilon}{\rho_l \log_e(D/d)}$$

$$\boxed{C' = \frac{2\pi\epsilon}{\log_e(D/d)}\ F/m}$$

This is the expression for capacitance/length for co-axial cable.

Energy Stored in Electric Field (Electro Static Energy):-

The amount of electrostatic energy stored can be found by calculating the work done in charging the capacitor.
we know,

$$\frac{dw}{dQ} = V \Rightarrow dw = V\,dQ$$

$$\Rightarrow dw = \frac{Q}{C}\,dQ \qquad \boxed{\because C = Q/V}$$

Thus, total work done is,

$$W = \int_0^{Q} \frac{Q}{C}\,dQ = \frac{1}{C}\int_0^{Q} Q\,dQ$$

$$= \frac{1}{C}\left[\frac{Q^2}{2}\right]_0^{Q}$$

$$\Rightarrow \boxed{W = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}Q.V = \frac{1}{2}CV^2}$$

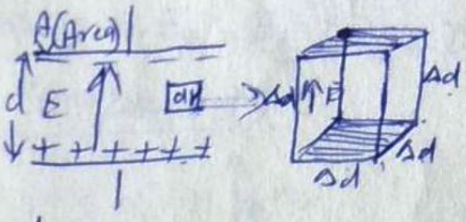
Energy Density in Static Electric Field :-

when a capacitor is charged to a potential diff. of V b/w plates, the energy stored is,

$$W = \frac{1}{2}CV^2 = \frac{1}{2}VQ$$

But in which part of the capacitor this energy is stored?
Energy is stored in the electric field b/w plates.

consider small cubical volume
$dv = (\Delta d)^3$ b/w the plates
where Δd = each side of small volume



If a thin sheet of metal foil are placed coincident with the top and bottom faces of the volume, the field will be undisturbed provided the sheets are sufficiently thin. The volume dv now consists of a small capacitor of capacitance,

$$\Delta C = \frac{\varepsilon(\Delta d)^2}{\Delta d} = \varepsilon\Delta d$$

Now, Potential difference ΔV of thin sheets is,

$$\Delta V = E\,\Delta d$$

Now, ΔW is the energy stored in volume dV i.e.

$$\Delta W = \frac{1}{2}\Delta C(\Delta V)^2 = \frac{1}{2}\,\varepsilon\,\Delta d\,(E\Delta d)^2$$

$$= \frac{1}{2}\varepsilon E^2\,dV$$

$$\Rightarrow \boxed{w' = \frac{\Delta W}{dV} = \frac{1}{2}\varepsilon E^2}$$ where w' = Energy density = $\frac{\text{Energy}}{\text{Volume}}$

The total energy stored by capacitor is given as.

$$\boxed{W = \int_v w' dV = \frac{1}{2}\int_v \varepsilon E^2 dv}$$

$$\boxed{W = \frac{1}{2}\int_v \vec{D}.E\,dv = \frac{1}{2}\int_v \frac{D^2}{\varepsilon}\,dv}$$ $[\because \vec{D} = \varepsilon\vec{E}]$

CONDITIONS AT BOUNDRY B/w TWO DIELECTRICS :-

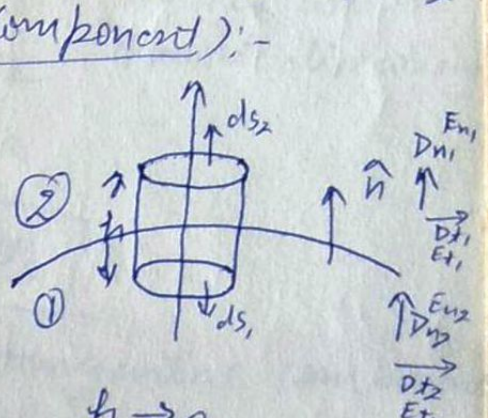
There are two boundary conditions such as,

(I) Normal components of $\vec{D}$ is continuous and that of $\vec{E}$ is discontinuous across the boundary i.e. $D_{n_1} = D_{n_2}$ & $E_{n_1} \neq E_{n_2}$

(II) Tangential components of $\vec{E}$ is continuous and that of $\vec{D}$ is discontinuous across the boundary i.e. $E_{t_1} = E_{t_2}$ & $D_{t_1} \neq D_{t_2}$.

(I) $D_{n_1} = D_{n_2}$ & $E_{n_1} \neq E_{n_2}$ (Normal Component):-

Let us imagine a disc enclosing a part of the boundary surface b/w two media as shown in fig.



Let ε_1 & ε_2 be ~~set~~ permittivity of two mediums. It is assumed that thickness of disc is very small. The only contribution to the outward flux ~~is~~ from the disc is from its flat surfaces

Then, by Gauss's law,

$$\int_V \nabla\cdot D\, dv = \int_V \rho\, dv = 0 \qquad \left[\because \text{no free charge at the surface}\right]$$

By divergence theorem,

$$\int_V (\nabla\cdot D)\, dv = \int_S \vec{D}\cdot dS = 0$$

$$\Rightarrow D_{n_1}\cdot d\vec{S_1} + \vec{D}_{n_2}\cdot dS_2 = 0$$

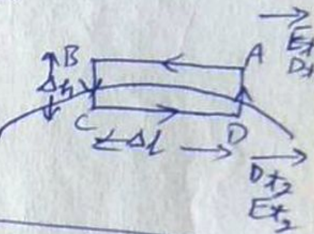
Now $dS_1 = -d\vec{S_2} \Rightarrow D_{n_1} dS_1 - D_{n_2} dS_1 = 0$

$$\Rightarrow \boxed{D_{n_1} = D_{n_2}} \Rightarrow \boxed{\varepsilon_1 E_{n_1} = \varepsilon_2 E_{n_2}}$$

But $\varepsilon_1 \neq \varepsilon_2 \Rightarrow \boxed{E_{n_1} \neq E_{n_2}}$

(ii) $E_{t_1} = E_{t_2}$ & $D_{t_1} \neq D_{t_2}$ (Tangential Component):-

Now, Consider a small closed rectangular path ABCDA of small length Δl & width Δh as shown in fig. Now work done by $\vec{E}$ around the closed path ABCDA is



$$\oint E\cdot dl = \int_A^B E\cdot dl + \int_B^C E\cdot dl + \int_C^D E\cdot dl + \int_D^A E\cdot dl = 0 \qquad \boxed{\because \oint E\cdot dl = 0}$$

Now consider $\Delta h \to 0$ i.e. BC & DA have length almost zero

$$\Rightarrow \int_A^B E\cdot dl + \int_C^D E\cdot dl = 0$$

$$\Rightarrow -E_{t_1}\cdot \Delta l + E_{t_2}\cdot \Delta l = 0 \qquad \left[\because AB \,\&\, E_{t_1} \text{ are in opposite direction}\right]$$

$$\Rightarrow \boxed{E_{t_1} = E_{t_2}} \Rightarrow \boxed{\frac{D_{t_1}}{\varepsilon_1} = \frac{D_{t_2}}{\varepsilon_2}}$$

But $\varepsilon_1 \neq \varepsilon_2 \Rightarrow \boxed{D_{t_1} \neq D_{t_2}}$

Hence Normal component of $\vec{D}$ & tangential component of $\vec{E}$ are continuous across the boundary while Normal component of $\vec{E}$ & Tangential component of $\vec{D}$ are discontinuous across the boundary.

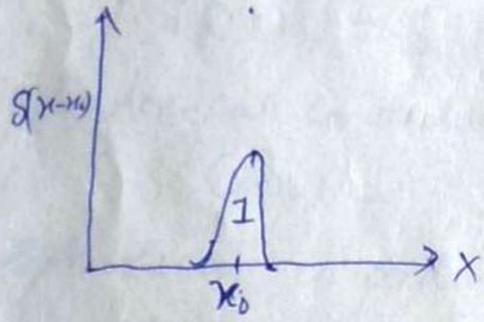
DIRAC DELTA :-

Dirac Delta, under the name of unit impulse has been widely used in circuit theory to represent a very short pulse of high amplitude.

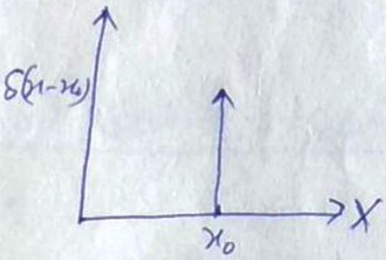
The Dirac delta at the point $x = x_0$ is represented by $\delta(x-x_0)$ and at $x=0$ it is denoted by $\delta(x)$ and can be given as

$$\int_a^b \delta(x-x_0) = 1 \quad \text{if } x_0 \text{ is in } (a,b)$$
$$= 0 \quad \text{if } x_0 \text{ is not in } (a,b)$$

It can be shown as,



Actual form



Approximate/symbolic form.

Another fundamental property is,

$$\int_a^b f(x)\,\delta(x-x_0)\,dx = f(x_0) \quad \text{If } x_0 \text{ is in } (a,b)$$
$$= 0 \quad \text{if } x_0 \text{ is not in } (a,b)$$

Now,

$$\int_a^b f(x)\,\delta'(x-x_0)\,dx = \left[f(x)\,\delta(x-x_0)\right]_a^b - \int_a^b f'(x)\,\delta(x-x_0)\,dx$$

$$\left[\because \left[f(x)\,\delta(x-x_0)\right]_a^b = 0\right]$$

$$= -f'(x_0) \quad \text{if } x_0 \text{ is in } (a,b)$$
$$= 0 \quad \text{if } x_0 \text{ is not in } (a,b)$$

llly

$$\int_a^b f(x)\,\delta^{(n)}(x-x_0)\,dx = (-1)^n f^{(n)}(x_0) \quad \text{if } x_0 \text{ is in } (a,b)$$
$$= 0 \quad \text{otherwise}$$

where (n) is nth order derivative.

The representation of a point charge or current element requires the use of 3-dimensional dirac delta. If r denotes the point (x, y, z), then 3-D delta at $r = r_0$ is $\delta(r - r_0)$ and at origin it is $\delta(r)$. Also,

$$\int_v \delta(r - r_0)\, dv = 1 \quad \text{if } r_0 \text{ is in } v.$$
$$= 0 \quad \text{if } r_0 \text{ is not in } v.$$

and

$$\int_v f(r)\, \delta(r - r_0)\, dv = 1 \quad \text{if } r_0 \text{ is in } v$$
$$= 0 \quad \text{otherwise}$$

In rectangular co-ordinates,

$$\delta(r - r_0) = \delta(x - x_0)\, \delta(y - y_0)\, \delta(z - z_0)$$

In cylinderical co-ordinates,

$$\delta(r - r_0) = \frac{\delta(\rho - \rho_0)\, \delta(\phi - \phi_0)\, \delta(z - z_0)}{\rho_0}$$

In spherical co-ordinates,

$$\delta(r - r_0) = \frac{\delta(r - r_0)\, \delta(\theta - \theta_0)\, \delta(\phi - \phi_0)}{r_0^2 \sin\theta_0}$$

Dirac Delta Representation For A Point Charge:-

It is convenient to write charge density ρ to denote a point charge. If a point charge q is located in vol. v, then,

$$\int_v \rho\, dv = q$$

If Point charge is outside v, then

$$\int_v \rho\, dv = 0$$

This can be shown if a charge q located at $r = r_0$ can be written as,

$$\rho(r) = q\, \delta(r - r_0)$$
$$= q\, \delta(x - x_0)\, \delta(y - y_0)\, \delta(z - z_0)$$

For a point charge at origin,

$$\rho(r) = q\,\delta(x)\,\delta(y)\,\delta(z)$$

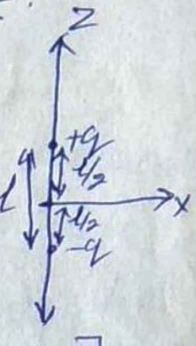
Line & sheet charge may be represented like,

$$\rho(r) = \rho_\ell\,\delta(x)\,\delta(y) \quad \text{where } \rho_\ell = \text{linear charge density along z-axis.}$$

and $\rho(r) = \rho_s\,\delta(z)$ where ρ_s = surface charge density along x-y plane.

Dirac Delta Representation for An Infinitesimal Dipole:-

Consider a small dipole along z-axis having length ℓ and charges $+q$ & $-q$ as shown in fig.



Now, charge density due to +ve & -ve charge can be given as,

$$\rho(r)^{+} = [+q\,\delta(x)\,\delta(y)\,\delta(z-\ell/2)] \;\&\; \rho(r)^{-} = [-q\,\delta(x)\,\delta(y)\,\delta(z+\tfrac{\ell}{2})]$$

$$\Rightarrow \rho(r) = \rho(r)^{+} + \rho(r)^{-} = q\,\delta(x)\,\delta(y)\,\delta(z-\ell/2) - q\,\delta(x)\,\delta(y)\,\delta(z+\ell/2)$$

$$= -q\,\delta(x)\,\delta(y)\,[\delta(z+\ell/2) - \delta(z-\ell/2)]$$

~~If~~ As length is very small for infinitesimal dipole i.e. $\ell \to 0$ and multiply & dividing by ℓ, we get

$$\rho(r) = \lim_{\ell\to 0}\left[-q\ell\,\delta(x)\,\delta(y)\left\{\frac{\delta(z+\ell/2) - \delta(z-\ell/2)}{\ell}\right\}\right]$$

$$\boxed{\rho(r) = -p\,\delta(x)\,\delta(y)\,\delta'(z)}$$

where $p = q\ell$ = Dipole moment

& $\delta'(z) = \lim_{\ell\to 0}\left[\dfrac{\delta(z+\ell/2) - \delta(z-\ell/2)}{\ell}\right]$ (By first principle)

This is the Dirac delta representation for an infinitesimal dipole.